Xinyuan Yan, Ph.D.

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EDUCATION

Postdoctoral Researcher, Neurosurgery & Neuroscience — April 2025 – Present
Baylor College of Medicine *Advisors: Benjamin Y. Hayden & Sameer A. Sheth*

Postdoctoral Researcher, Psychiatry & Neuroscience — March 2022 – March 2025
University of Minnesota *Advisors: Alexander Herman & David Darrow*

Ph.D., Cognitive Neuroscience — September 2016 – January 2022
IDG/McGovern Institute for Brain Research, State Key Laboratory of Cognitive Neuroscience and Learning
Beijing Normal University

B.Sc., Psychology — September 2012 – June 2016
Northeast Normal University

FELLOWSHIPS & AWARDS

Travel Award, Computational and Systems Neuroscience (Cosyne), Montreal — 2023
Trainee Professional Development Award, Society for Neuroscience — 2022
MnDrive Neuromodulation Research Fellowship, University of Minnesota Medical School — 2022
Fellowship, Computational and Cognitive Neuroscience (CCN) Summer School, CSHL Asia — 2021
Fellowship, Kavli Summer Institute in Cognitive Neuroscience, UC Santa Barbara — 2019
Distinguished Graduate Research Fellowship, State Key Laboratory of Cognitive Neuroscience and Learning — 2018
National Graduate Research Fellowship, China — 2018

PUBLICATIONS

*† indicates co-last author. * indicates co-first author.*

Concepts, language, single-neuron recordings, and hippocampus

1. **Yan X**, Chavez AG, Franch M, …, Hayden BY†, Sheth SA† (2026). Shared neural geometries for bilingual semantic representations in human hippocampal neurons. ***Cell*.**
2. **Yan X**, Li JA, Franch M, . . . Hayden BY†, Sheth SA† (2026). Polysemanticity in human hippocampal neurons. ***bioRxiv*.**
3. Chericoni A, Diao C, **Yan X**, . . . Sheth SA†, Hayden BY† (2026). Neural geometry in the human hippocampus enables generalization across spatial position and gaze. ***Neuron*.**
4. Chavez AG, **Yan X**, Franch M, . . . Hayden BY (2026). Hippocampus serves as a repository for spoken and heard word meanings during conversations. ***Nature Communications (in press)*.**
5. Xiong HD*, **Yan X***, Li JA, Xue J, Mattar MG, Wilson RC (2026). Thinking Under Uncertainty: Evidence Use and Information-Seeking in Language Models. ***arXiv preprint*.**

6. Ismail T, Chavez AG, **Yan X**, ... Hayden BY (2026). Estimation of neuronal tuning for word meaning from passively recorded naturalistic speech. ***bioRxiv*.**

Concepts about uncertainty and implications for affective disorders

Skill set: computational modeling, intracranial EEG

7. **Yan X**, Peterson CM, Schmidt LM, Koenig S, Nixdorf DR, Herman A, Darrow D (2026). Impaired adaptive learning in chronic pain contributes to apathy. ***eLife***, 2026-01.

8. **Yan X**, Ebitz RB, Grissom N, Darrow DP, Herman AB (2025). Distinct computational mechanisms of uncertainty processing explain opposing exploratory behaviors in anxiety and apathy. ***Biological Psychiatry: Cognitive Neuroscience and Neuroimaging.***

9. **Yan X**, Ebitz BR, Darrow DP, Herman AB (2025). Nonlinear modulation of human exploration by distinct sources of uncertainty. ***bioRxiv***, 2025-04.

10. **Yan X**, König SD, Ebitz RB, Hayden BY, Darrow DP, Herman AB (2025). Dynamic prefrontal coupling coordinates adaptive decision-making. ***Research Square***, rs-3.

11. Knep E, **Yan X**, Chen CS, Jacob S, Darrow DP, Ebitz RB, . . . Herman AB (2025). Social aloofness is associated with non-social explore–exploit decisions. ***Communications Psychology***, 3(1), 106.

12. Cao D*, Cui J*, **Yan X***, Zhang X, Dang Y, Zhao H, . . . Zhang Y (2025). Dynamic interactions in human amygdala subregions support hierarchical processing of fearful faces. ***bioRxiv***, 2025-11.

13. Jin T, Zhang S, **Yan X**, Vilares I (2026). Social Influence in Donation Behavior: The Effect of Mean, Variance, and Psychopathy. ***eLife***, 2026-08.

Social-Concepts & Social Cognition

Skill set: computational modeling, fMRI

14. **Yan X**, Yong X, Huang W, Ma Y (2018). Placebo treatment facilitates social trust and approach behavior. ***PNAS***, 115(22), 5732–5737.

15. Liu Y, Li S#, Lin W#, Li W#, **Yan X**, Wang X, Pan X, Rutledge RB, Ma Y (2019). Amygdala activity tracks deviations from social preferences. ***Nature Neuroscience***, 22(4), 633–641. [# co-second author]

16. Wang D, **Yan X**, Li M, Ma Y (2017). Neural substrates underlying oxytocin effect: A quantitative meta-analysis of pharmaco-imaging studies. ***Social Cognitive and Affective Neuroscience***, 12(10), 1565–1573.

Self-Concepts and implications for major depressive disorder

Skill set: fMRI

17. **Yan X** (2024). Neural dynamics of self-referential processing and the insight for decoding self-concepts. ***Journal of Neuroscience***, 44(30). [Commentary]

18. Feng C†, **Yan X**†, Huang W, Han S, Ma Y (2018). Neural representations of the multidimensional self in the cortical midline structures. ***NeuroImage,*** 183, 291–299. [† co-first author]

Cortisol and the stress model

Skill set: resting-state fMRI

19. Feng Z, Zhang H, Zhou T, Yu X, Zhang Y, **Yan X*** (2023). Dynamic functional connectivity changes associated with psychiatric traits and cognitive deficits in Cushing's disease. ***Translational Psychiatry***, 13(1), 308. [* corresponding author]
20. Shang G, Zhou T, **Yan X**, . . . Zhang Y (2025). Multiscale analysis reveals hippocampal subfield vulnerabilities to chronic cortisol overexposure: Evidence from Cushing's disease. ***Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*.**
21. Liu B, Mao Z, **Yan X**, Yang H, Xu J, Feng Z, Zhang Y, Yu X (2024). Structural network topologies are associated with deep brain stimulation outcomes in Meige syndrome. ***Neurotherapeutics***, 21(4), e00367.

EDITORIAL SERVICE

- **Guest Associate Editor**, *Frontiers in Human Neuroscience* — Research Topic: *Methods in Cognitive Neuroscience: 2023* (Oct 2023 – Dec 2023)

AD-HOC REVIEWER

PNAS
Nature Communications
Communications Biology
Translational Psychiatry
Brain and Behavior
Psychoradiology (Oxford Academic) · Self and Identity · Frontiers in Psychology · Frontiers in Psychiatry · Frontiers in Human Neuroscience · Frontiers in Endocrinology · Frontiers in Robotics and AI · BMC Neuroscience.

PROFESSIONAL MEMBERSHIPS

- Society for Neuroscience
- Society of Biological Psychiatry (Regular Member)

CONFERENCE PRESENTATIONS

Oral Presentations, Invited Talks

- **Yan X.**, *Stable meaning, flexible codes: how the human hippocampus builds and reuses semantic geometry.* NYU Langone Health (self invited), August 5th, 2026, New York, USA.
- **Yan X.**, *Shared neural geometries for bilingual semantic representations in human hippocampal neurons.* NYU Psychology (invited by Prof. David Poeppel), August 6th, 2026, New York, USA.
- **Yan X.**, Koneig S, Ebitz B, Hayden, B, Darrow D, Herman A. *Dynamic prefrontal coupling coordinates adaptive decision-making*, Society for Neuroscience, November 15-19, 2025, San Diego, USA.
- **Yan, X.**, König, S., Ebitz, B., Hayden, B., Darrow, D., & Herman, A. (2024). *Deciphering neural mechanisms of explore–exploit decision-making in the prefrontal cortex*. Computational Psychiatry Conference, Minneapolis, USA.
- **Yan, X.**, Ebitz, B., . . . Darrow, D., & Herman, A. (2022). *The behavioral profile of the exploration–exploitation dilemma and its implications for anhedonia and anxiety*. Computational Psychiatry, New York City, USA.

Poster Presentations

- **Yan X**, *Shared neural geometries for bilingual semantic and grammatical representations in human hippocampal neurons*. 9th annual conference on cognitive computational neuroscience. 2026, August 6th, New York, USA.
- **Yan, X.**, König, S., Ebitz, B., Darrow, D., & Herman, A. (2023). *Intracranial electrophysiological evidence for a novel neuro-computational mechanism of exploration–exploitation decision-making in humans*. Computational and Systems Neuroscience (Cosyne), Montreal, Canada.
- **Yan, X.**, König, S., Ebitz, B., Darrow, D., & Herman, A. (2022). *Intracranial electrophysiological evidence for a novel neuro-computational mechanism of exploration–exploitation decision-making in humans*. Society for Neuroscience Annual Meeting, San Diego, USA.

ADDITIONAL TRAINING

- Computational and Cognitive Neuroscience (CCN) Summer School, *Cold Spring Harbor Laboratory*, 2021
- Online Summer School for Computational Neuroscience, *Neuromatch Academy*, 2020
- Kavli Summer Institute in Cognitive Neuroscience, *UC Santa Barbara*, 2019
- Coursera coursework (2020): Computer Simulations · Social Network Analysis · Network Dynamics of Social Behavior · Julia Scientific Programming · Facial Expression Recognition with Keras · Introduction to HTML5

TEACHING & MEDIA

Mentoring junior graduate students in core skills for scientific research:

- Representational similarity analysis
- Resting-state fMRI analysis
- Diffusion decision model & value-based DDM
- Reinforcement learning & Markov decision processes
- Attractor networks & decision-making
- Neural networks-basic concepts
- Building a personal academic website (HTML5 + CSS3)
- Science Communication-https://www.youtube.com/watch?v=ywhGMM80lt0

GRADUATE STUDENTS/TRAINEES SUPERVISED:

- Ana G. Chavez (5th year PhD student, Hayden Lab)
- Assia Chericoni (5th year PhD student, Hayden Lab)
- Taha Ismail (Research technician, Hayden & Sheth Lab)
- Will Baltazar (MD student, Sheth Lab)
- Siddharth Marathe (MD-PhD student, Sheth Lab)
- James L Belanger (Research technician, Hayden Lab)
- Minye Zhou (1st year PhD student, Sheth Lab)

References:

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Dr. Jay Hennig, Assistant Professor
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Dr. Alexander Herman, Assistant Professor
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